A Current-efficient Wideband Cellular RF Receiver for Multi-Band Inter- and Intra-Band Carrier Aggregation using 14nm FinFET CMOS

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***Abstract*—A wideband cellular RF receiver for multi-band carrier aggregation employing a current-efficient wideband low noise amplifier and a frequency-band switchable transformer is demonstrated in a 14nm FinFET CMOS technology. The proposed wideband low-noise amplifier can support multiple-channel RF signals for intra-band carrier aggregation with high performance and low DC current consumption. Moreover, the frequency-band switchable transformer is used to support a size-efficient receiver up to 5 carrier components carrier aggregation. The receiver operates at frequencies between 0.6 to 2.7 GHz. The receiver has conversion gain more than 62 dB and noise figure less than 5 dB at all carrier aggregation combinations.**

***Index Terms*—carrier aggregation (CA), FinFET CMOS, low-noise amplifier (LNA), receiver, transformer (XFMR)**

I. Introduction

The recent demonstration of a receiver (RX) front-end architecture demands a wideband receiver RFIC to support multiple cellular bands with minimized RX input ports, as opposed to the narrow-band receiver RFIC requiring so many input ports and external matching network, dedicated to each band. Moreover, the highly-integrated LNA-PA Module with integrated duplexers (LPAMid) is currently used in front of the wideband receiver RFIC for better sensitivity. This provides additional design challenges with respect to achieving a high linearity and current-efficiency in the wideband receiver.

Moreover, in order to achieve higher data rate in the recent trend of mobile communication, three or more adjacent or non-adjacent radio frequency (RF) carrier allocations within the band can be aggregated to increase the effective signal bandwidth. This downlink carrier aggregation (CA) technique was introduced in 3GPP LTE Rel-10 to meet the demand to improve data transmission speed [1]. Recently, the Rel-14 [2] defines up to five carrier components (CC) with non-contiguous intra-band CA which fragments RF channels of the same band.

These face new RF RX design challenges. Foremost, since all intra-band CCs within the same band need to share single RX input port, it is very important to develop a RX architecture which can separate the received intra-band RF signals into different baseband signals efficiently

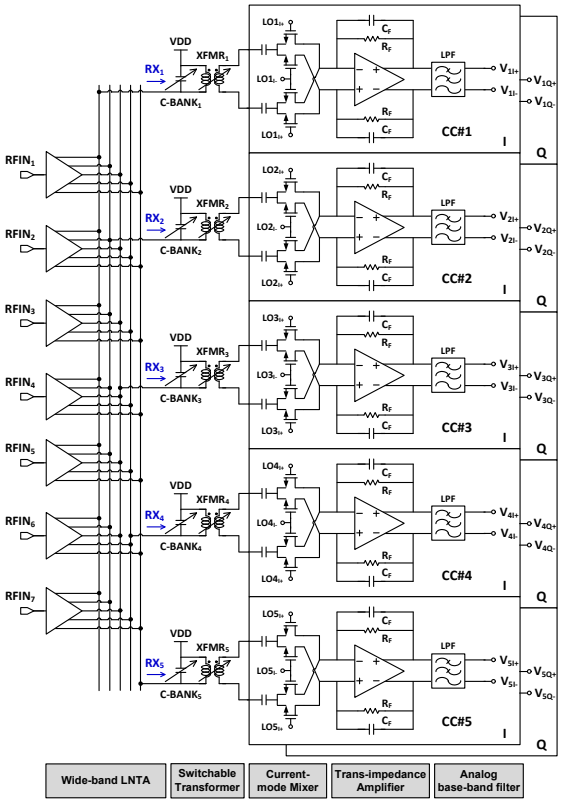


Fig. 1. A proposed RX architecture supporting up to 5CC CA.

without performance degradation compared to single CC operation. Moreover, the received intra-band RF signals may have unequal power levels due to communication with different base stations or frequency-dependent fading effect. To resolve this, the RF receiver supporting intra-band CA should provide the independent RF gain control at each CC to achieve the high dynamic range. Besides, since an external LNA as LPAMid shows a wideband operation through a single output port, subsequent RX RFIC should also achieve the RX performance with good input matching, high linearity and relatively low noise over a wide frequency range with low power consumption. In recent studies, several state-of-art receivers supporting intra-band CA have been published, showing superior

receiver performance [1][3]-[6]. However, intra-band signal processing approaches in either analog [3] or digital domain [4] could suffer from various blocker scenarios due to its low-IF operation with a single LO. Another approach, which utilizes N-path filter with high-order or multi-channel band-pass filtering profile from the reconfigurable trans-impedance amplifier (TIA) feedback path, provides the extra noise contribution and nonlinearities [5]. Furthermore, the aforementioned approaches for intra-band CA have no function to control LNA gain for each CC independently to achieve the similar front-end dynamic range to the nominal 1CC operation. On the other hands, the receiver architectures in [1][6] show a straight-forward approach integrating 3 receivers with allocated LOs for each CC and support the independent RF gain control of multi-path LNAs for intra-band CA. However, multi-path LNA architectures are designed only for narrowband operation with high Q input impedance, which cannot support wideband operation with LPAMid and the LNA structure in [6] requires relatively high DC current consumption to compensate the performance degradation in multiple intra-band CA mode in spite of 3CC intra-band CA.

In this work, the new receiver architecture using a current-efficient 2-stage wideband LNA is proposed to fully support multiple CCs inter- and intra-band CA scenarios, which is implemented in 14 nm CMOS FinFET (Fin Field Effect Transistor) process. The proposed wideband LNA topology maintains the receiver performance even for multiple CCs intra-band CA up to 5CC with low current consumption. Moreover, a modified derivative superposition (DS) method is applied to the proposed LNA for linearity improvement.

II. Receiver Design

The block diagram of the proposed receiver is shown in Fig.1. The designed receiver consists of 7 single-ended wideband LNAs with 7 input ports, 5 sets of transformers (XFMR) based on band-switchable balun and 5 identical down-conversion mixers followed by TIAs and analog base-band filters (ABBs) to support multiple CCs CA operations up to 5CC. The 7 single-ended LNAs have connections to all of 5 XFMRs for independent 5CC inter- and intra-band CA operations.

Fig.2 (a) and (b) show the simplified circuit schematics, which can be a candidate for the wideband LNA architecture supporting up to 5CC intra-band CA. This topology is mainly realized with parallel-connected cascade transconductance (g_m) cells, and each CC path as RX_1 to RX_5 has a dedicated sliced-g_m cell for each CC operation. Common-gate (CG)-FETs are used to control the gain of each path independently through g_m current steering. The wideband input return loss in Fig.2 (a) and (b) can be achieved by parallel RC feedback and input Q-

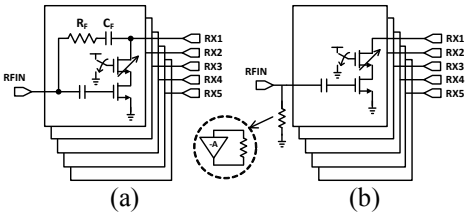


Fig. 2. A conventional wideband LNA for intra-band CA with a) parallel RC feedback and b) input Q-degradation

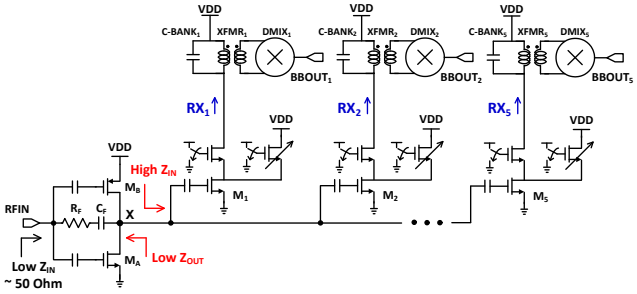


Fig. 3. A proposed wideband LNA for current-efficient intra-band CA.

factor degradation, respectively. However, in Fig. 2 (a) using the parallel RC feedback, when multiple CCs intra-band CA operation is activated specifically, 5CC with 5 slice-g_m cells, its input impedance becomes smaller which results in gain degradation in spite of 5 times higher DC current consumption. Therefore, this approach requires multiple times more DC current to have identical gain to 1CC operation. Moreover, this approach needs complicated multi-band calibrations for gain and gain step of LNA at each intra-band CC case, since the loop gain is not constant over a wide frequency range. In Fig. 2 (b) using the input Q degradation technique, the required DC current of the LNA also increases in direct portion as the number of intra-band CC increases. Moreover, the noise figure (NF) of this approach suffers from input resistor noise contribution. To avoid NF degradation given by the input shunt resistor, this resistor can be replaced with an amplifier with a resistive feedback. However, this causes linearity degradation as well as additional DC current.

Fig.3 shows the detailed circuit schematic of the proposed wideband LNA architecture for current efficient multiple CCs intra-band CA operations. The proposed LNA is realized with 2-stage amplifiers, which consists of a CMOS (M_A and M_B) amplifier with the parallel RC feedback and subsequent 5 multi-path cascode-FET amplifiers to activate each RX path (from RX_1 to RX_5) as 1st-and 2nd-stage, respectively. The CMOS amplifier with the RC feedback can not only produce voltage gain at node X but also generate wideband input matching for wideband receiver operation. Moreover, it can provide low source impedance to 2nd-stage cascode-FET amplifiers. Each CS-FET (M_1, M_2, M_3, M_4, and M_5,) of the 2nd-stage

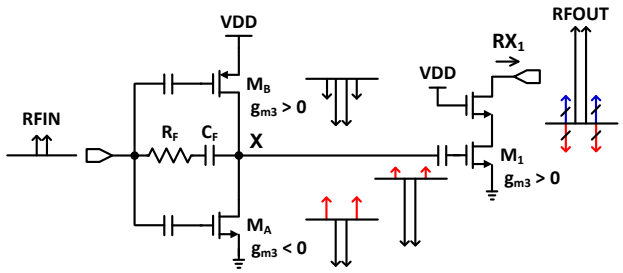


Fig. 4. IMD3 compensation concept of the proposed LNA

amplifier can be independently selected to decide which XMFR and down-conversion mixer (DMIX) to be used for 1CC operation. In the multiple CCs intra-band CA mode, 2nd-stage cascade FETs are selectively turned-on to activate dedicated RX paths while sharing the 1st-stage CMOS amplifier (e.g. while M_A and M_B are turned-on, M_1 and M_2 can be only turned-on in 2CC intra-band CA and all CS-FETs need to be turned-on in 5CC intra-band CA). The CC1 RX gain determined by voltage gain of 1st-stage CMOS and g_m current of assigned 2nd-stage FET is not affected by on/off conditions of other 2nd-stage FETs, since the voltage gain at node X is maintained by the output impedance of 1st-stage CMOS LNA with the RC feedback, which is much smaller than input impedance of turned-on 2nd-stage CS-FET. Moreover, RF input matching condition is fixed by the 1st-stage CMOS LNA with the RC feedback, different from Fig. 2(a). This means that multiple CCs intra-band CA operation can be realized only with additional activation of assigned 2nd-stage cascode–FETs with small current consumption increase. Therefore, a current-efficient multiple CCs intra-band CA operation can be realized with only small additional DC current from 2nd-stage cascode-FETs turn-on and without any receiver performance degradation of gain, NF, and linearity compared to 1CC operation. Moreover, the CG-FETs of each 2nd-stage LNA path offer independent gain controllability through g_m current steering without wideband gain step calibration of LNA, different from Fig. 2(b).

As shown in Fig. 4, a modified DS method to achieve IIP3 improvement over a wide frequency range is also developed and applied to the proposed 2-stage wideband LNA, which is less sensitive to the process variation than a conventional optimum biasing technique. In a 1st-stage CMOS amplifier, a complementary DS method with CS configuration is applied; NMOS and PMOS are operated as main and auxiliary transistors, which are biased at strong and weak inversion, respectively. This makes the total g_m increase and second-order interaction (caused by second-order distortion combined with feedback) decrease owing to the same signs of g_m and second-order transconductance (g_{m2}) in NMOS and PMOS. Moreover, since third-order transconductance (g_{m3}) of NMOS and PMOS have different signs due to different operation

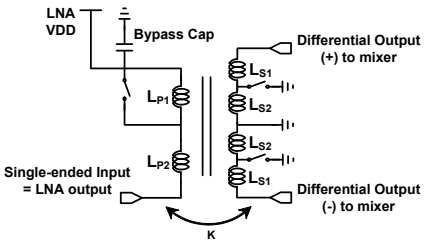


Fig. 5. A proposed band-selective balun transformer

region, negative g_{m3} of NMOS is compensated with positive g_{m3} of PMOS. However, since positive and negative characteristic of g_{m3} in each FET are not symmetric, the remaining negative g_{m3} in 1st-stage CMOS amplifier output is subsequently compensated with 2nd-stage CS-FET which is biased at weak inversion with positive g_{m3} characteristic to widen IMD3 cancellation window. As a result, as shown in Fig. 4, the final third-order distortion of the proposed wideband 2-stage LNA can be improved within a finite bias-voltage range instead of just a point as conventional one. A size efficient band-switchable transformer is also applied to realize a single to differential conversion between LNA and I/Q differential mixer [1]. Its operating frequency can be adjusted from 0.5 to 3 GHz for size efficient 5CC inter- and intra-band CA by changing the inductance of both primary and secondary side, as shown in Fig. 5.

III. Measurement Results

The designed receiver has been implemented in a 14 nm FinFET CMOS technology and characterized with a 1 V supply. Fig. 6 shows a chip microphotograph, whose area is 3.3 mm^2 including 7 LNAs, 5 sets of down-conversion paths (XFMR, DMIX, TIA, and ABB) and 5CC LNA output routings. The fully integrated receiver includes 5 identical down-conversion paths with independent frequency synthesizers, enabling also the 5CC inter-band CA. Two down-conversion paths driven by single frequency synthesizer is also possible to support up to 4×4 MIMO application with single LO. Receiver performance of proposed receiver is measured in 1CC, 3CC, and 5CC intra-band CA modes. 1CC operation for RX_1 is realized only with turned-on M_A, M_B, and M_1 (M_2 to M_5 are remained-off). 3CC intra-band CA operation is realized with additionally turned-on M_2 and M_3 as RX_2 and RX_3, respectively, and 5CC intra-band CA operation is realized with all FET turn-on. Fig. 7 plots the measured NF and total LNA current consumption at 1 V supply for RF frequencies from low-band (600 MHz) to high-band (2700 MHz) in 1CC, 3CC, and 5CC intra-band CA modes, showing RF-to-BB conversion gain of 62 dB achieved. The measured NF at all cellular frequency is less than 5 dB at all CA cases without external matching. Compared to 1CC operation, the degradation of NF and gain in 5CC intra-band CA is less than 0.5 dB and 1 dB, respectively.

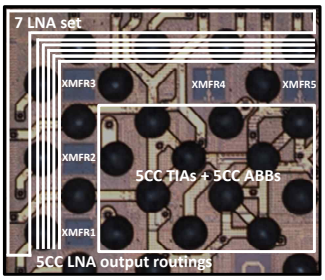
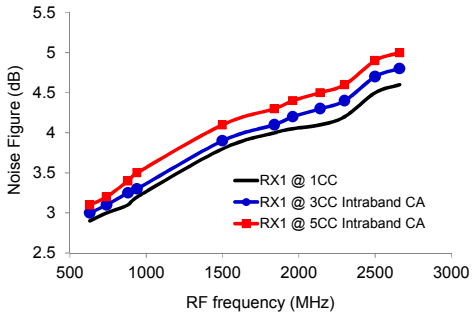
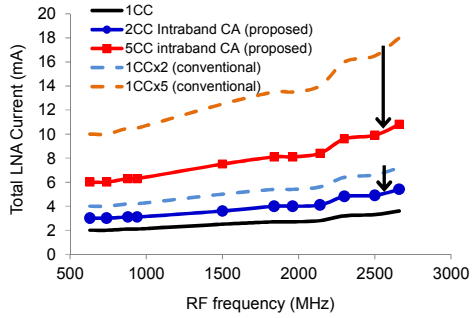


Fig. 6. Fabricated chip photograph of proposed receiver



(a)



(b)

Fig. 7. For receiver conversion gain of 62 dB, (a) measured noise figure at 1CC and 3CC/5CC intra-band CA; (b) measured total LNA current at 1CC and 3CC/5CC intra-band CA.

It is noticed that additional LNA current consumptions for further intra-band CC activation is less than 1.5 mA which is only contributed by 2nd-stage CS-FET. Therefore, performance of gain and NF at all RX paths can be maintained in all multiple-CCs intra-band CA modes with small current consumption. The linearity performance is also checked by measuring the base-band output fundamental and inter-modulated tones to verify the applied modified-DS method. The third-order linearity performance is checked through two-tone tests at B5, B2, and B7. Fig. 8 shows the measured in-band IIP3 with two tones at 12.5 MHz and 24 MHz offset frequency from LO and the measured out-band IIP3 with two tones at transmitter (TX) frequency and TX frequency-duplex spacing, respectively. The obtained IIP3 are higher than -2.5 dBm and there is no linearity degradation from multiple CCs intra-band CA. Table I shows performance of proposed receiver and compare it with that of prior arts.

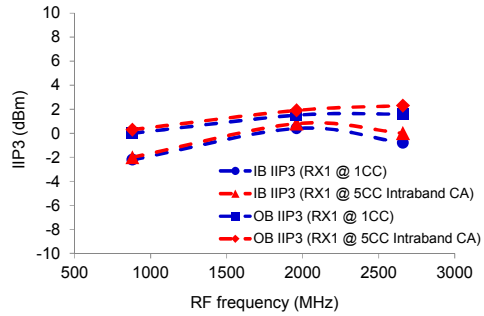


Fig. 8. Measured in-band IIP3 and out-band IIP3 at 1CC and 5CC intra-band CA.

TABLE I
PERFORMANCE COMPARISON OF INTRA-BAND CA RECEIVER

	[1]	[3]	[4]	[5]	[6]	**This work**
Tech/Supply	28nm/1.2V	65nm/1.5V	45nm/1V	45nm/1V	65nm/1.2V	14nm/1V
Standard	GSM /WCDMA /LTE	LTE	LTE	LTE	WLAN	GSM /WCDMA /LTE
Max. CC	3	2	3	3	3	5
Power (mW)	19	155	15	96	52	16
NF (dB)	< 3.5	4.5	3.9	6	5	5
Gain (dB)	70	45	37	50	55	62
IIP3 (dBm)	-12.5	-15	-13.7	-20	-7.5	-2.5

IV. CONCLUSION

In this paper, a complete LTE-advanced wideband cellular receiver for 5CC inter- and intra-band CA is presented. The receiver is realized with 7 wideband 2-stage LNAs and 5 down-conversion paths including dedicated size-efficient XFMRs. The proposed wideband LNA topology shows the current-efficient reception of 5 intra-band RF signals without performance degradation and good linearity employing the modified-DS method.

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